One graph. Three stages.

From verified constraints to dynamic diffusion guidance and constraint-aware output repair

Mixed-type tabular data | Frozen TabDiff backbone | Research-oriented method

1 **Construct**
What relations matter?

2 **Guide**
When / where to act?

3 **Repair**
What remains missing?

FRAMING · 02

One question, three challenges

> **How can we apply the right constraints to the right rows at the right diffusion step — and still produce a realistic final table?**

WHICH RULE?

Not every pattern is a rule

Some constraints apply only to certain people or situations; others are statistical tendencies.

WHEN?

Rows become readable at different times

A correction that helps one row may be premature or unnecessary for another.

FINAL TABLE

Do we keep the right groups?

Even if every selected row looks valid, we may keep **too many rows from one group** and too few from another.

Example: 10% of training rows are students, but only 2% remain in the final synthetic table.

FRAMING · 03

Three problems, one graph

GRAPH CONSTRUCTION

1 · Semantic mismatch

A plausible pattern may have exceptions or insufficient evidence.

Need: verified, applicable constraints.

GRAPH GUIDANCE

2 · Temporal mismatch

A fixed constraint penalty cannot know whether a value is already readable or still changing.

Need: row- and step-aware activation.

OUTPUT REPAIR

3 · Set mismatch

Selecting rows for good overall statistics can still miss constraint-specific subgroup patterns.

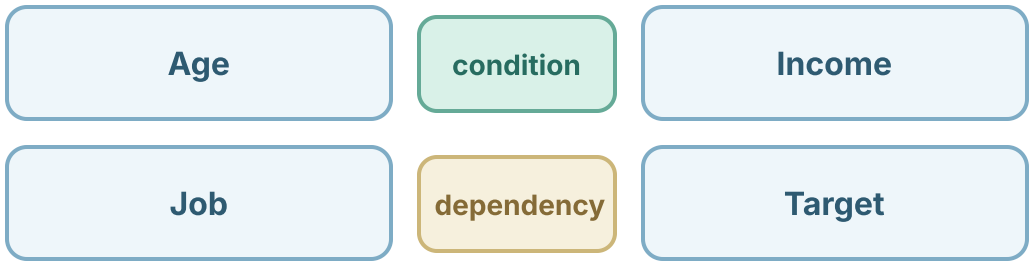
Need: graph-aware set repair.

The same graph should define **what to preserve**, **when to act**, and **what to check at the output**.

FRAMING · 04

What makes the graph “dynamic”?

A fixed set of candidate constraints



We **do not** claim to learn a new graph topology at every step.

A changing active graph

	Early	Middle	Late
Row A	Off	On	Off
Row B	Off	N/A	N/A
Row C	Off	On	Off

Each row and timestep has its own **active subset of constraints**.

Dynamic graph = dynamic constraint activation, not continuous graph discovery.

FRAMING · 05

Three contributions · a single dynamic graph

C1 · Construct a usable constraint graph

Represent constraints with their columns, conditions, reliability and allowed actions. This gives the controller a **consistent graph to use**.

C2 · Activate and inject the graph dynamically

At every relevant reverse step, use the current row state to decide **which constraints are active now**; apply bounded, type-aware corrections.

C3 · Repair graph distributions at the output

After initial distribution selection, reuse graph-defined subgroups to **detect remaining gaps** and make a few safe membership swaps.

One graph links three decisions: **what is valid → when to intervene → what is missing in the final set**.

RELATED WORK · 06

Closest graph-based tabular generation work

EMNLP 2025 · ACCEPTED

SPADA — LLM-induced dependency graph

Builds a sparse dependency graph from LLM knowledge, then generates columns conditioned on graph parents.

Key comparison: the graph guides the *feature generation order/dependencies*; it is not a per-row, per-timestep constraint activation controller inside diffusion.

Yang et al. · paper ↗

ACM CODASPY 2026 · ACCEPTED

MoE-T — dependency graph-gated generation

Uses dependency-graph gating in a mixture-of-experts model for tabular generation and functional dependency preservation.

Key comparison: graph-controlled generation, but not the proposed *row × diffusion-time active constraint graph*.

Dhooghe et al. · accepted list ↗

Prior work makes **graph-guided generation** a meaningful comparison; our question is **how its active constraints should change during denoising**.

Closest constraint injection in tabular diffusion

2026 · PREPRINT

SCTab-Diff — semantic priors in denoising

Discovers intra-column semantics and inter-column symbolic rules with LLM assistance, validates them on training data, and conditions diffusion denoising on these priors.

Key comparison: priors are injected into denoising; our focus is **which factor is active for which row, at which state/step**, plus reuse at output repair.

Wang et al. · arXiv ↗

ICLR 2026 · ACCEPTED

HARPOON — inference-time constraint guidance

Uses manifold-aware guidance for diverse inference-time conditions, including inequality constraints, on tabular diffusion.

Key comparison: a strong guidance baseline; our hypothesis concerns **constraint-specific, row/state/time-dependent activation** and graph-shared output selection.

Shankar et al. · ICLR ↗

The proposed distinction is a testable design hypothesis, not a claim that existing models cannot respond to input conditions.

NOVELTY · 08

Gap: one fixed correction does not fit every row or step

EXAMPLE · STATISTICAL TENDENCY

Age ↔ marital status

Older people may be more likely to be married in the training data.

But this is **not** a rule that every older person must be married. A fixed, strong correction applied to every row would distort valid single/divorced cases.

OUR DESIGN

Activate only when justified

For each row and step, ask:

1. Is this relation applicable?
2. Are its columns readable yet?
3. Is any correction still needed?
4. Is the intervention safe?

A valid exception can stay unchanged.

Two mismatches of fixed guidance

Across rows

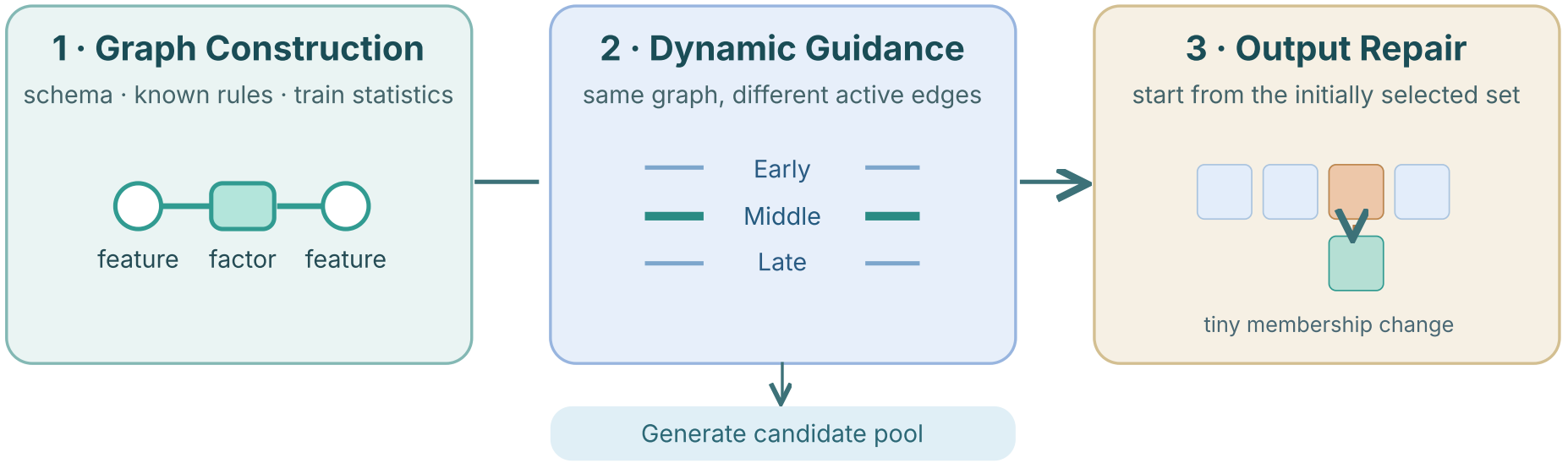
Different ages, categories and exceptions imply different applicable relations.

Across diffusion steps

An early noisy estimate can trigger an incorrect correction; later it may already be satisfied.

METHOD · 09

Method overview · three stages



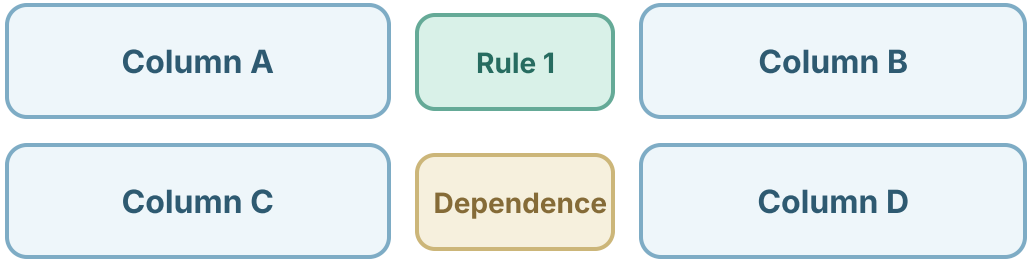
Construction defines constraints; guidance makes them dynamic; repair preserves them in the final table.

Why a constraint graph?

A table has relations—not isolated columns

- A hard rule may involve several columns.
- A conditional rule only applies to some rows.
- A statistical relation may be useful but should not become a strict rule.

Graph = columns + constraints



The constraint node can connect one, two, or more columns.

> “Constraint graph” is the intuitive name; “factor graph” describes the typed mathematical representation.

METHOD · STAGE 1 · 11

Construct verified, usable constraints

1 · CANDIDATES

Discover

Known domain rules, metadata and relations learned from training data.

LLM suggestions may be included offline.

2 · VERIFICATION

Check

Check types, where rules apply, support and counterexamples.

3 · PERMISSION

Assign use

Safe to enforce · use only for scoring · observe · reject.

Example distinction

“Sum must match total” → enforce if verified.

“Income correlates with age” → statistical relation, not a hard law.

This stage supports dynamic guidance. The paper does not need to claim a new constraint-mining algorithm.

Dynamic graph: active constraints vary

Row A / Row B / Row C

	Early	Middle	Late
Row A	Wait	Act	Stop
Row B	Wait	N/A	N/A
Row C	Wait	Act	Stop

Conceptual example; the actual pattern must be measured from trajectory traces.

What changes?

- **Row:** rule applies here, but not elsewhere.
- **Time:** relevant columns become readable.
- **State:** a satisfied relation may no longer need correction.
- **Safety:** uncertain or costly actions can be skipped.

The active graph is **row-specific, step-specific and state-dependent**.

When does a constraint become active?

Active(e, row, step) = Applicable × Readable × Needed × Safe

Applicable

The condition holds for this row.

Readable

Relevant values can be assessed at this step.

Needed

The current sample has a gap worth correcting.

Safe

Rule is trusted; intervention stays within budget.

Simplified from the report's full six-part activation rule. No separate learned gating network is claimed.

How do active constraints act?

PROJECTION

Hard numerical rule

Move values minimally toward the valid region.

Example: an inequality or valid range

TRANSPORT

Statistical relation

Adjust numerical distributions or conditional dependencies.

Example: age-conditioned income distribution

LOGIT MESSAGE

Categorical relation

Adjust probabilities of still-masked categories.

Example: category conditioned on a row attribute

Different constraint types need different actions—but the **same active graph** decides whether to use them.

Why intervene in the middle?

EARLY	MIDDLE	LATE
Preview is noisy	Readable + room to improve	Little correction budget

Reverse diffusion →

Weak, targeted pulses

The controller acts only when a relation is eligible and the current state still needs correction.

Avoid overcorrection

Strong or continuous guidance can improve one constraint while distorting fidelity or coverage.

Timing claim comes from earlier sweeps; early/middle/late/continuous ablation remains part of final confirmation.

Set mismatch: good rows ≠ good table

Training table

A factor-conditioned subgroup



Target proportions within a subgroup.

Selected synthetic table

The same subgroup after selection



All rows may satisfy hard rules, but this subgroup distribution differs.

Set mismatch is **factor-conditioned distribution mismatch**, not simply “invalid rows”.

Numbers above are illustrative, not measurements from Adult/Shoppers.

Stage 3 starts with a good set of selected rows

STEP 1 · DISTRIBUTION SELECTION

Choose a representative subset

Generate a **larger pool**, e.g. 8,192 candidate rows, then keep 4,096 rows that better match training distributions (individual columns, column pairs, common/rare groups).

We call this the **distribution selection** step. It only selects rows; it does not edit cell values.

STEP 2 · GRAPH CHECK & REPAIR

Keep the important constraint groups

A subset may look good overall, yet misrepresent a group described by our constraint graph.

Ours checks the selected set using the same graph as Stage 2, and **swaps a few rows** only when needed—without sacrificing overall quality.

Stage 3 does not run diffusion again: **choose rows → check graph distributions → make a few swaps or keep the set unchanged**.

Stage 3: a small correction to the selected set

1 · GENERATE

Larger pool

Generate more rows than required so there are alternatives to choose from.

2 · SELECT

Good overall table

Keep a subset close to train marginals, pairs and coverage.

3 · GRAPH CHECK

Look for missing groups

Compare constraint-specific groups and relations in train vs selected data.

4 · REPAIR IF NEEDED

Swap very few rows

Replace a few selected rows using candidates still in the pool; otherwise do nothing.

The same constraint graph guides row-level corrections during diffusion and **set-level checks** after generation.

METHOD · 19

What is actually shared across stages?

THE SAME CONSTRAINTS

During diffusion

Should a relation act on this row now?

⇄

At the final set

Is the relation still represented across selected rows?

Shared meaning

Same endpoints, applicability, relation type and authority.
This links local guidance to set diagnostics.

Different decisions

Trajectory: change estimated values/logits.
Output: change membership only.

One graph, two decision scales. That is the unifying novelty—not three unrelated tricks.

Compared methods and what the metrics mean

Compared methods

Slide name	Meaning
TabDiff	Original frozen generator
Ours · Guidance	Dynamic constraint injection only
Random subset	Random rows from an enlarged generated pool
Ours · Selection	Guidance + distribution selection (before graph repair)
Ours · Full	Guidance + selection + graph repair

How to interpret quality metrics

Metric	Meaning
Shape error ↓	Difference in single-column distributions
Trend error ↓	Difference in relationships between columns
Alpha ↑	Precision-like sample fidelity / realism
Beta ↑	Recall-like coverage of real-data modes
TSTR AUC ↑	Downstream task performance of synthetic-trained model
Graph gap ↓	Mismatch on graph-defined relations/groups; internal audit

Shape/Trend are shown as errors (smaller is better) in Adult/Shoppers/Magic/Default; on News, the historical score form is shown with ↑. Do not compare those scales directly.

Dataset coverage: where has Ours worked?

Dataset	Best available stage	Shape	Trend	What the evidence says
Adult	**Ours · Full**	**0.860949 ↓**	**2.017464 ↓**	Strongest final case; **4/5** local records
Shoppers	**Ours · Full**	1.102046 ↓	1.803424 ↓	Graph alignment/coverage improved; not best vs prior selection on fidelity
Magic	Ours · Full (boundary)	0.956562 ↓	2.334173 ↓	Graph improvement, but Trend advantage not robust
Default	Ours · Full (boundary)	0.964895 ↓	1.435227 ↓	Graph improves; external quality does not improve vs selection
Diabetes	Ours · Selection only	1.353019 ↓	8.165711 ↓	Selection confirmed; graph repair was not feasible
News	Ours · Guidance only	97.3812 ↑	91.5203 ↑	Guidance improvements on separate score-based News protocol

Source: report snapshot (22 Oct 2026). Protocols and stages differ; table is a coverage map, **not** a single cross-dataset SOTA ranking. Final results: 3 fresh/held-out seeds as reported; News is a separate paired evaluation.

News: dynamic constraint guidance (no selection)

News · 3 paired seeds · direct reported means

Score-based evaluation; larger Shape/Trend is better

Metric	TabDiff	Ours · Guidance	Interpretation
Shape score ↑	97.3387	**97.3812**	Single-column fidelity
Trend score ↑	91.4938	**91.5203**	Cross-column relations
Alpha ↑	94.4515	**96.4019**	Realism / precision-like
Beta ↑	42.2843	**42.3378**	Mode coverage
TSTR R^2 ↑	−3.0396	**−2.8857**	Regression utility
TSTR NRMSE ↓	2.0063	**1.9216**	Regression error

This compares TabDiff with dynamic guidance **before** output selection; utility remains weak in absolute terms (negative R^2).

Adult: final three-stage method

Adult · 3 fresh seeds · same final confirmation

Shape/Trend: error ↓

Metric	Ours · Guidance*	Ours · Selection*	Ours · Full
Shape error ↓	1.232297	0.869082	**0.860949**
Trend error ↓	2.539276	2.041732	**2.017464**
Alpha ↑	—	—	**99.068206**
Beta ↑	—	—	**48.155382**
TSTR AUC ↑	—	—	**0.897593**

Full method gives the lowest Shape/Trend error among these paired variants; **4/5 local baseline records**. Stage 3 changes selected rows, not cell values.

* Guidance/selection means reconstructed algebraically from report-provided paired differences (not directly tabulated absolutes). Missing cells were not reported.

Final results across datasets: direct metrics

Full-method confirmation · same-batch comparison

Shape ↓ = better marginals; Trend ↓ = better column relationships

Dataset	Method stage	Shape error ↓	Trend error ↓	Interpretation
Adult	Ours · Guidance*	1.232297	2.539276	**Full best** on both
	Ours · Selection*	0.869082	2.041732	
	Ours · Full	**0.860949**	**2.017464**	
Shoppers	Ours · Guidance*	1.491208	2.037819	Selection has lower errors; Full improves graph consistency
	Ours · Selection*	**1.077901**	**1.773058**	
	Ours · Full	1.102046	1.803424	
Magic	Ours · Guidance*	1.202843	**2.300632**	Better graph score; not robust Trend gain
	Ours · Selection*	0.965420	2.341414	
	Ours · Full	**0.956562**	2.334173	
Default	Ours · Guidance*	1.341219	1.462354	Full does not outperform selection
	Ours · Selection*	**0.957794**	**1.435182**	
	Ours · Full	0.964895	1.435227	

* Guidance/selection: values reconstructed from paired differences. Three-seed means; graph consistency and coverage are evaluated separately.

Shoppers: local baselines and Ours (screen)

Shoppers · earlier single-run screening protocol

Not final confirmation

Method	Shape error ↓	Trend error ↓	Alpha ↑	Beta ↑	TSTR AUC ↑
TabDiff	1.6044	2.5302	96.3738	**50.3320**	0.8978
TabSyn	1.6652	1.9929	99.0039	46.3363	0.8817
ForestDiffusion	14.5687	4.6032	91.0807	49.7168	0.8992
Ours · Projection + selection (screen)	**1.0493**	**1.5375**	**99.0390**	49.4466	**0.9038**

Ours refers here to an **earlier projection + distribution-selection variant**, not the final graph-repair method. Values are historical local protocol only.

Magic: local baselines and Ours (screen)

Magic · earlier one-run screen / historical baseline context

Not final confirmation

Method	Shape error ↓	Trend error ↓	Alpha ↑	Beta ↑	TSTR AUC ↑
TabDiff	1.6857	1.3771	**98.9840**	47.3242	0.9181
ForestDiffusion	3.3541	3.6141	—	**54.0137**	0.9092
Ours · Projection guidance	1.190401	3.314354	98.254423	48.829753	**0.929434**
Ours · Projection + selection (pair-aware)	**0.983095**	**0.777460**	98.489471	49.344076	0.924755
Ours · Projection + selection (marginal)	0.986567	0.811485	98.853886	49.907227	0.926762

Pair-aware vs marginal are **distribution-selection variants** from an earlier screen, not different final graph-repair configurations.

News: which guidance component helps?

News · one-run controlled variants (different from 3-seed News)

Metric directions shown in headers

Variant	Shape ↑	Trend ↑	Alpha ↑	Beta ↑	TSTR R^2 ↑	NRMSE ↓
TabDiff	97.4853	91.1478	94.8853	42.3690	−0.8374	1.3555
Ours · Hard-rule correction only	97.5503	91.1495	96.7654	42.3958	−0.7999	1.3416
Ours · Row-conditional transport only	97.5043	91.1564	94.9472	42.3958	−0.6665	1.2909
Ours · Global transport only	97.5057	91.1552	94.9505	42.3576	−0.8672	1.3664
Ours · Full + row-conditional transport	97.5601	**91.1568**	96.9119	42.3958	−0.5489	1.2446
Ours · Full + global transport	**97.5613**	91.1559	**96.9200**	**42.4512**	**−0.5438**	**1.2425**

This is a trajectory-only factorial. Global and row-conditional full variants are close; it does **not** establish universal dominance for dynamic conditioning.

Adult: categorical guidance diagnostic

Adult · older, separate 3-seed diagnostic experiment

Not final Ours method

Metric	TabDiff	Ours · Categorical-only variant	Better direction
Shape error	**1.2211**	1.2357	↓
Trend error	**2.5383**	2.6021	↓
Alpha	98.6041	**98.7113**	↑
Beta	**48.7126**	48.4744	↑
TSTR AUC	0.89719	**0.89820**	↑

A promising one-seed categorical setting did not generalize across three seeds. This is a **diagnostic ablation**, not evidence against the newer final graph-repair setting.

Why Stage 3 uses small, conditional swaps

START

Good selected table

First choose rows that reproduce overall training distributions.

PROBLEM

Graph-specific gap

Some constraint-conditioned groups may still be over- or under-represented.

AVOID

Replace everything

Reselecting a whole table may damage fidelity even when graph scores improve.

OURS

Few safe swaps

Keep original quality, swap very few rows, skip repair if no gap remains.

The method alters **which rows are retained**. It never creates new values in Stage 3.

Ablation coverage · method-linked experiment map

GRAPH

Stage 1

Sources, trust, graph size, conflicts and topology.

GUIDANCE

Stage 2

Row × time activation, stability, scheduling, actions.

REPAIR

Stage 3

Distribution selection, shared constraints, sparse swaps and gate.

EVALUATION

Entire method

Fair baselines, seeds, subgroups, privacy, cost.

SYSTEM

Efficiency

Pool ratio, diffusion steps, GPU profile, memory.

A separate interactive HTML map lists the full candidate ablations, controls, metrics, costs and priorities for discussion.

A · Is graph construction meaningful?

Arm	Question	Main readout
No graph	What can the frozen backbone do?	External validity + quality
Semantic rules only	What do verified domain constraints add?	Applicable violation rate
Statistical relations only	Is distribution knowledge enough?	Conditional fidelity
Verified mixed graph	Does combining factor families help?	Validity + fidelity Pareto
Unverified / low-support proposals	Why must we check authority?	False action + exception loss

Datasets: Adult + News; optionally a focused LLM/no-LLM discovery analysis, not a full new research direction.

B · Does the graph need to be dynamic?

Arm	What changes?	Hypothesis
Fixed/global	Same factors across rows/steps	Static guidance is too coarse
Row-aware only	Factor applicability changes by row	Avoid irrelevant corrections
Time-aware only	Readability changes by timestep	Avoid premature interventions
Row + time	Both decisions combined	Selective guidance gives best trade-off
No residual / no uncertainty gate	Drop necessity or safety check	Identify the useful part of dynamics

Fair comparison

Same factor set & action operators.

Match guidance budget / total movement.

Report validity–fidelity Pareto, not one tuned point.

C · Does the intervention design matter?

Action family

Arm	Tests
Projection only	Hard rule correction
Numerical transport only	Distribution correction
Categorical messages only	Category dependencies
Full typed actions	Complementarity

When & how strong?

Arm	Tests
Early / middle / late	Best moment to intervene
One pulse / continuous	Overcorrection
Trust bound on / off	Safety of movement
Weak / locked / strong	Pareto sensitivity

Choose dataset by relevant factor types; do not run categorical ablation on numerical-only Magic.

Ablation: why do graph-based output swaps help?

Variant	What does it test?
Ours · Guidance only	Generation without selection or repair
Random selection	Does a larger pool alone explain the gain?
Ours · Distribution selection	Strong set-level baseline without graph-specific repair
Ours · Add graph penalty	Can simple weighted graph loss work?
Ours · Reselect entire set	Why is a small number of swaps safer?
Ours · Graph-first swaps	What happens if graph improvement comes first?
Ours · Quality-first swaps	Does preserving distribution quality help?
Ours · Full + skip-if-consistent	Does abstaining avoid unnecessary changes?

Use exactly the same generated pool for selection-only ablations; compare Adult/Shoppers first.

E · Can the method be trusted?

REPRODUCIBILITY

Robustness

Adult/Shoppers: ≥5 final seeds if feasible.

Show paired gains, confidence intervals and no-op rate.

QUALITY

Coverage & constraints

Rare subgroup recall, conditional coverage, independent violations, noisy-factor stress.

PRACTICALITY

Cost & selection budget

Pool ratio 1×/2×/4×; swaps 0/0.25/0.5/1%; runtime, memory, DCR.

A small internal graph loss is not enough: external metrics and compute fairness decide the claim.